

# EchoCode

## Business Revenue Plan | Cohort and Campus

Voice-first AI pair programming for people who read code they did not write

|                                                           |                                           |                                           |                                         |
|-----------------------------------------------------------|-------------------------------------------|-------------------------------------------|-----------------------------------------|
| <b>\$0.13</b><br>cost per question<br>(60-s idle timeout) | <b>81%</b><br>gross margin<br>cohort deal | <b>62%</b><br>gross margin<br>campus deal | <b>\$440k</b><br>gross profit<br>year 3 |
|-----------------------------------------------------------|-------------------------------------------|-------------------------------------------|-----------------------------------------|

## Executive summary

EchoCode sells to institutions, not to students. Students get a permanent free tier and act as the distribution channel; the invoice goes to the organisation whose measured outcome improves. Two customer types carry the plan. **Bootcamps** buy per cohort, decide in weeks, and are scored on completion rate. **University departments** buy per seat per year, decide in months, and represent the larger contract.

The model is built on a verified cost base rather than an assumption. AssemblyAI's Voice Agent API is billed per second of session duration at \$4.50/hour, and code-card generation through the LLM Gateway costs \$0.0014 per card – under 1% of the cost of goods sold. Cost of goods is therefore almost entirely voice session time, and it is metered per install.

**The central insight.** The cost driver is idle session time, not question volume. A question consumes 45 seconds of speech but bills up to three minutes of open session. Reducing the idle timeout from three minutes to sixty seconds takes cost per question from \$0.225 to \$0.131 – a single constant worth more to gross margin than any change to price.

## Target customers

|                     | Cohort (bootcamps)                 | Campus (departments)              |
|---------------------|------------------------------------|-----------------------------------|
| <b>Buyer</b>        | Programme director                 | Department head or course lead    |
| <b>Unit sold</b>    | Per learner, per cohort            | Per seat, per academic year       |
| <b>Price</b>        | \$25 per learner                   | \$36 per seat                     |
| <b>Typical deal</b> | 30 learners = \$750                | 500 seats = \$18,000              |
| <b>Sales cycle</b>  | 2-6 weeks                          | 3-9 months                        |
| <b>Their KPI</b>    | Completion rate                    | Pass rate, teaching-hours load    |
| <b>Role in plan</b> | Beachhead: revenue now, proof fast | Scale: larger contracts, renewals |

Bootcamps come first. They decide quickly, their economics are transparent, and dropout is an event they can already price. University departments are the larger prize but cannot be closed inside a first commercial year, so they are sold into course-by-course rather than campus-wide to bypass procurement.

## Pricing architecture

Allowances are metered per install, today in seconds of question and answer audio; moving them to **session minutes**, the unit AssemblyAI actually bills, is planned. Allowances are pooled across a cohort or department so that pricing reflects the cohort average instead of the heaviest individual user.

| Tier    | Price                | Included                   | Max COGS    | Margin      |
|---------|----------------------|----------------------------|-------------|-------------|
| Free    | \$0                  | 10 questions / month       | \$1.31 / mo | Acquisition |
| Cohort  | \$25 / learner       | ~36 questions / 12 weeks   | \$4.73      | 81%         |
| Campus  | \$36 / seat / yr     | 120,000 session-min pooled | \$13.78     | 62%         |
| Pro     | \$20 / month         | 60 questions / month       | \$7.88      | 61%         |
| Overage | \$0.15 / session-min | Beyond pooled allowance    | \$0.075     | 50%         |

## Unit economics

| Cost component          | Rate            | Basis                                       |
|-------------------------|-----------------|---------------------------------------------|
| Voice Agent session     | \$0.075 / min   | Per second of open session (\$4.50/hr)      |
| Code card (LLM Gateway) | \$0.0014 / card | qwen3.5-4b-32k-fast, ~12k tokens in         |
| Infrastructure          | ~\$360 / year   | Vercel Pro and Upstash Redis, all customers |

### Cost per question, by idle timeout

A question is 45 seconds of speech. The rest of the billed time is the session sitting open while the developer reads and thinks, plus the idle tail before it closes.

| Idle timeout       | Billed min / question | Cost / question | Status                     |
|--------------------|-----------------------|-----------------|----------------------------|
| 3 minutes          | 3.00                  | \$0.225         | As shipped today           |
| 60 seconds         | 1.75                  | <b>\$0.131</b>  | <b>Planning assumption</b> |
| 30 seconds         | 1.25                  | \$0.094         | Later option               |
| Audio only (floor) | 0.75                  | \$0.056         | Theoretical minimum        |

**One idle timeout for every plan.** Sixty seconds on Free, Pro, Cohort and Campus alike. Tiering it by plan is a later option, and we will model it only once real usage data exists: a three-minute Pro timeout, for example, would cost \$13.50 of a \$20 price at full use, a 33% margin.

## Deal-level profit and loss

### Cohort deal - one bootcamp intake of 30 learners

| Line         | Calculation              | Amount          |
|--------------|--------------------------|-----------------|
| Revenue      | 30 learners x \$25       | \$750.00        |
| Voice COGS   | 1,080 questions x \$0.13 | (\$141.75)      |
| Code cards   | ~324 cards x \$0.0014    | (\$0.45)        |
| Gross profit | 81% margin               | <b>\$607.80</b> |

A bootcamp partner running four intakes a year is worth **\$3,000 in revenue and \$2,431 in gross profit**, won with roughly five hours of founder-led selling.

### Campus deal - 500 seats at \$36

| Line         | Calculation               | Amount          |
|--------------|---------------------------|-----------------|
| Revenue      | 500 seats x \$36          | \$18,000        |
| Voice COGS   | 52,500 questions x \$0.13 | (\$6,891)       |
| Code cards   | ~15,750 cards x \$0.0014  | (\$22)          |
| Gross profit | 62% margin                | <b>\$11,087</b> |

## Adoption sensitivity - the risk is success

Blended questions per seat equal the adoption rate multiplied by the usage of those who adopt. Because voice is a real cost of goods, high adoption compresses margin. This is the question an experienced judge or investor asks first, and the pooled allowance with paid overage is the answer to it.

| Adoption       | Blended q / seat / yr | Voice COGS     | Gross profit    | Margin     |
|----------------|-----------------------|----------------|-----------------|------------|
| 40% - light    | 70                    | \$4,594        | \$13,406        | 74%        |
| 60% - expected | 105                   | <b>\$6,891</b> | <b>\$11,109</b> | <b>62%</b> |
| 80% - mandated | 160                   | \$10,500       | \$7,500         | 42%        |
| 100% - heavy   | 240                   | \$15,750       | \$2,250         | 13%        |

**Downside is contractually capped.** The 500-seat licence includes 120,000 pooled session minutes. Beyond that, usage bills at \$0.15 per session minute - double our cost. The heavy-adoption row therefore does not erode margin in practice; it converts into expansion revenue, and the deals that grow fastest are the ones that grow without a new sale.

Prices and deal counts in this document are modelled hypotheses, not signed contracts. The cost base - \$0.075 per session minute and \$0.0014 per code card - is verified against live AssemblyAI pricing.

## Three-year projection

Bottom-up from deal counts, at the \$0.13 planning cost per question (60-second idle timeout). Free-tier consumption is carried as a marketing cost at 15% monthly-active of cumulative sign-ups.

|                       | Year 1          | Year 2           | Year 3           |
|-----------------------|-----------------|------------------|------------------|
| Bootcamp partners     | 10              | 30               | 60               |
| Cohorts delivered     | 40              | 120              | 240              |
| Campus contracts      | 2               | 8                | 25               |
| Pro subscribers       | 50              | 200              | 600              |
| Cohort revenue        | \$30,000        | \$90,000         | \$180,000        |
| Campus revenue        | \$36,000        | \$144,000        | \$450,000        |
| Pro revenue           | \$12,000        | \$48,000         | \$144,000        |
| Total revenue         | <b>\$78,000</b> | <b>\$282,000</b> | <b>\$774,000</b> |
| Voice COGS            | (\$24,176)      | (\$91,035)       | (\$262,986)      |
| Free tier (marketing) | (\$4,725)       | (\$23,625)       | (\$70,875)       |
| Gross profit          | <b>\$49,099</b> | <b>\$167,340</b> | <b>\$440,139</b> |
| Gross margin          | <b>63%</b>      | <b>59%</b>       | <b>57%</b>       |

Margin declines by design: campus is lower-margin than cohort and grows as a share of the mix. Figures are gross profit and exclude salaries and sales costs. At today's 3-minute timeout (\$0.225 a question) the same model gives Year-1 gross profit of about \$28,500 (36%), which is why the timeout change comes first.

**The AssemblyAI credits fund year one.** \$5,000 is 66,000 session minutes: about 22,000 questions today and 38,000 after the timeout change, covering our year-one free tier at \$4,725 of modelled consumption.

## Risks and mitigations

| Risk                                   | Mitigation                                                                            |
|----------------------------------------|---------------------------------------------------------------------------------------|
| High adoption compresses campus margin | Pooled allowance plus \$0.15/min overage; a 60-second idle timeout cuts unit cost 42% |
| Campus sales cycles run 3-9 months     | Bootcamps carry year-one revenue; campus sold course by course, avoiding procurement  |
| GitHub Copilot is free for students    | Different job, different buyer: Copilot writes code, we explain code already written  |
| Voice is a genuine cost of goods       | Allowances metered per install; margin is 57-81% and we say so                        |

## What must be validated next

- 1. Questions per student per week.** Every figure in this plan hangs off this number. One teaching term of usage logs settles it, and the product already meters usage per install.
- 2. One real price quote.** A single bootcamp or lecturer stating what they would pay per learner is worth more than any projection on these pages.
- 3. Adoption rate per seat.** Determines whether a campus deal earns 74% or 13%, and therefore whether the overage term is commercial protection or the main revenue line.